PRINTED CIRCUIT BOARD · AUDIO POWER

Class-AB Audio Amplifier

Complete analogue design flow — schematic capture through to a fabrication-ready board.

AT A GLANCE

End-to-end class-AB power amplifier, taken from first-principles circuit design to a routed, manufacturable PCB.

TOOLCHAIN

- KiCad — schematic & layout
- LTSpice — pre-layout SPICE
- Gerber / drill export

DESIGN TARGETS

Topology	Class-AB
Verify	SPICE-led
Routing	Analogue-aware
Ground	Copper pour
Output	Gerber + drill

WHY CLASS-AB

Biasing the complementary output pair slightly into conduction suppresses class-B crossover distortion without the standing dissipation of class-A.

// OVERVIEW

This project covered the full design of a class-AB audio power amplifier, taken from first-principles circuit design through to a routed, manufacturable printed circuit board. The objective was hands-on command of the professional PCB flow: schematic capture, simulation-led verification, sourceable component selection, footprint assignment, analogue-aware board layout, and the generation of fabrication outputs.

// DESIGN RATIONALE

A class-AB output stage was chosen as the deliberate compromise between the linearity of class-A and the efficiency of class-B. Biasing the complementary output devices slightly into conduction substantially suppresses the crossover distortion of a pure class-B stage while avoiding the high quiescent dissipation of class-A. This demanded attention to bias-network stability and to the thermal behaviour of the output devices.

// IMPLEMENTATION

Schematic capture	The input stage, voltage-gain stage, bias network and complementary output stage were captured in KiCad with hierarchical organisation and consistent net naming.
Simulation	The signal path was modelled in LTSpice before layout to verify gain, bandwidth and output biasing, and to confirm crossover behaviour ahead of committing to a physical design.
Components & footprints	Real, sourceable parts were selected for every position with appropriate voltage, current and power ratings; each symbol was given a verified footprint checked against manufacturer land-pattern recommendations.
Board layout	Deliberate analogue routing — short direct audio traces, small-signal/power separation, considered grounding, and a copper pour for a low-impedance return and improved thermal spreading.
Manufacturing outputs	The board passed KiCad's design-rule checks; industry-standard Gerber and drill files were exported, leaving the design ready for any commercial fab house.

// OUTCOMES & SKILLS DEMONSTRATED

The project produced a complete, fabrication-ready amplifier design and a working command of the full schematic-to-Gerber workflow. Competencies: analogue circuit design, SPICE-based pre-layout verification, footprint and library management, analogue PCB layout discipline (signal-path routing, grounding, copper pours), and preparation of manufacturing documentation to industry standards.